



JUBAID HOSSEN

Hardware and Firmware Design Engineer

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ACADEMIC CREDENTIALS

Bachelor of Science in Electrical and Electronic Engineering

Jashore University of Science and Technology (JUST) [↗](#)

CGPA: 3.37/4.00

Jan 2018 – Jan 2023

Jashore, Khulna, Bangladesh

WORK EXPERIENCE

Walton Digi-Tech Industries Ltd. (1.5 Year) [↗](#)

Jul 2024 – Ongoing | Chondra, Gazipur

Firmware and Hardware Design Engineer

Electronics Research & Innovation(R&I)

Key Responsibilities:

- Hardware design for **power electronics converters, inverters**, and various home appliance products.
- Embedded systems architecture planning and real-time firmware development for power electronics applications.

TECHNICAL SKILLS

Power Electronics & Hardware

- DC-DC converters, inverters, UPS systems
- Analog and digital circuit design
- MATLAB, LTspice, Proteus

Embedded Systems & Microcontrollers

- MCU: STM32, ESP32, PIC MPU: NVIDIA Jetson Nano
- Frameworks: ESP-IDF, STM32-HAL, LVGL [↗](#), The Yocto Project

PCB Design

- Multi-layer PCB design
- EMI/EMC-aware layout practices
- Altium Designer, EasyEDA

Interfaces & Communication

- UART, SPI, I²C
- USB (CDC / custom class hardware support)
- ADC/DAC signal conditioning

DEVELOPED PRODUCTS

Uninterrupted Power Supply (UPS) [↗](#)

Jul 2024 – Mar 2025

Model name: Walton arc | UX01, 800 VA/480W

Role: Firmware and Hardware Designer

Brief:

- Modified square wave, 4 MOSFET H-Bridge inverter design
- Bootstrap-type high-side MOSFET driver with integrated shoot-through protection
- DMA-based ADC sampling with precision timing
- Controlled by STM32F103C8T6 MCU.

Rechargeable Fan [↗](#)

Aug 2024 – Feb 2025

Model name: Walton W17OA-AS (17")

Role: Hardware Designer

Brief:

- PWM-based speed control through MOSFET
- Ultra-low power consumption through careful component selection and deep sleep mode
- MCU and MOSFET-based BMS
- Controlled by Chinese APM32 MCU

PROJECTS

High Frequency DC-DC Converter

Aug 2025 – Nov 2025

500W

Role: Hardware Developer

Brief: Collaborative research project on high-frequency DC-DC converters for inverter front-end design

Supervised by: Md. Tarequzzaman , Lecturer, Dept. of EEE, JUST

String Fault Detection System on PV Power Plant

Sep 2024 – Dec 2024

Role: Hardware and Firmware Developer

Brief: The project was funded by the **Mongla 100 MW (Energon) Solar Power Plant** to detect PV faults on their solar arrays.

- Used ESP32 for wireless mesh network communications and fault detection.
- The system is connected to IoT via Thingspeak.

Supervised by: Dr. Engr. Emran Khan , Associate Professor and Chairman, Dept of EEE, JUST

Development of a Prototype-based Low-Budget Ventilator for COVID-19 Patients.

Oct 2021 – Feb 2023

Role: Firmware Developer

Brief: This project was funded and supervised by Professor Md. Amzad Hossain, Dr. Engr. Postdoc  to aid coronavirus-affected patients by providing a low-budget ventilator.

- Designed the servo-based airflow control mechanism.
- Ported firmware to **FreeRTOS**, improving real-time control performance and task scheduling.
- Participated in hospital testing and co-authored in publication (ICETSD-25).

VOLUNTEER EXPERIENCE

General Member

Shohayok Foundation, Community Welfare & Education Club

- Teaching basic primary education to underprivileged children
- Supporting financial assistance for patients requiring critical medical treatment.
- Organizing free medical camps and medicine distribution for low-income communities

General Member

JUST Robotics Society

- Gained hands-on experience in robotics, programming, and electronics.
- Contributed to team projects involving robot design, assembly, and testing.
- Supported the club in preparing for and participating in robotics competitions.
- Assisted in organizing workshops, seminars, and demonstrations.

REFERENCES

Md. Azmol Haque, *First Senior Additional Director, Research & Innovation(R&I)*, Walton Digi-Tech Industries Ltd.
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Dr. Engr. Imran Khan, *Associate Professor and Chairman, Department of EEE*, Jashore University of Science & Technology
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